

Product Requirements Document (PRD)

Field	Details
Project Title	MEDIVISTA AI — Multimodal Medical Imaging AI Copilot for Grounded Analysis, Report Generation and Clinical Decision Support
Student Name	Deepen
Roll No	202410700023
Year & Section	3rd Year – Semester 5
Project Type	Multimodal AI / Deep Learning / Medical Computer Vision / Vision-Language System
Primary Domain	Medical Imaging, Computer Vision, Natural Language Processing, Multimodal AI and Clinical Decision Support
Supported Imaging Modalities	X-Ray, CT, MRI and Ultrasound
Initial Research Modality	Chest X-Ray
Primary Dataset	MIMIC-CXR for the initial Chest X-Ray research pipeline
Additional Modality Datasets	Modality-specific de-identified research datasets for CT, MRI and Ultrasound, selected according to availability, licensing, annotations and project feasibility
Baseline Architecture	CNN + LSTM with Visual Attention for the initial Chest X-Ray report-generation benchmark
Proposed Research Direction	Modality-Specific Medical Vision Models + Abnormality Detection + Anatomical Grounding + Localization / Segmentation + Measurement + Structured Clinical Representation + Uncertainty Estimation + Evidence-Grounded Reasoning + Grounded Report Generation + Clinical Decision Support

Field	Details
Core Product Concept	AI Evidence Chain: Image → Finding → Anatomical Location → Measurement → Structured Finding → Report Statement → Clinical Evidence → Clinical Decision Support
Application Type	Multimodal Medical Imaging Analysis, Grounded Report Generation and Clinical Decision-Support Research Platform
Interface	React-based Medical Imaging AI Workstation
Backend / AI Services	FastAPI-based Backend + Python / PyTorch AI Services
Database	PostgreSQL / Research Database
Project Duration	12 Weeks
Prototype Status	Educational / Academic Research Prototype

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Project Proposal

1.1 Overview

1.1 Overview

Radiology is a critical component of modern healthcare, where medical imaging plays a central role in the detection, characterization and monitoring of diseases. Different imaging modalities such as X-Ray, Computed Tomography (CT), Magnetic Resonance Imaging (MRI) and Ultrasound provide complementary information about anatomical structures, pathological findings and disease progression. Interpreting these examinations requires visual understanding, anatomical localization, clinical context, uncertainty assessment and communication of findings through structured medical reports.

This project proposes MEDIVISTA AI — a Multimodal Medical Imaging AI Copilot for Grounded Analysis, Report Generation and Clinical Decision Support.

The system is designed as a research-oriented multimodal medical imaging platform capable of processing X-Ray, CT, MRI and Ultrasound studies through modality-specific AI pipelines. Instead of treating all medical images as a single homogeneous input type, the architecture uses specialized vision models appropriate to each modality while maintaining a common structured clinical representation and reasoning framework.

The system will process an uploaded medical imaging study through a modality-specific analysis pipeline. Depending on the imaging modality, the system will identify clinically relevant findings, localize or segment abnormalities where appropriate, extract anatomical information and measurements, estimate prediction confidence and uncertainty, and convert the visual observations into a structured clinical representation.

A central concept of the system is the AI Evidence Chain:

Image → Finding → Anatomical Region → Measurement → Structured Finding → Report Statement → Clinical Evidence → Clinical Decision Support

This evidence chain is intended to maintain traceability between the original medical image and the clinical information generated by the AI system. The system will therefore not treat report generation as an isolated image-to-text task. Instead, generated clinical statements will be linked to detected findings, anatomical evidence, model confidence and supporting clinical information whenever available.

The structured clinical representation will be provided to a multimodal reasoning component responsible for generating a grounded radiology report containing structured Findings and Impression sections. The report-generation process will incorporate visual evidence, detected abnormalities, anatomical context, measurements and uncertainty information.

Beyond report generation, MEDIVISTA AI will include an evidence-grounded Clinical Decision Support layer. This layer will use structured imaging findings together with retrieved clinical evidence to provide contextual diagnostic and management-oriented information. A Retrieval-Augmented Generation (RAG) approach may be used to retrieve relevant medical evidence and connect recommendations to supporting sources.

The system will also support longitudinal comparison where appropriate, allowing current imaging findings to be compared with previous studies to identify changes, progression or stability. In addition, a treatment-assistance component may generate a clinician-reviewable draft of possible management considerations based on the available evidence. Such outputs will remain decision-support information and will not constitute autonomous prescribing or final treatment decisions.

The primary research implementation will initially use the MIMIC-CXR dataset for the Chest X-Ray pipeline because it provides paired de-identified images and radiology reports suitable for image-report research. Additional datasets for CT, MRI and Ultrasound will be selected separately according to modality, annotation availability, licensing, de-identification requirements and project feasibility.

The research will begin with a CNN-LSTM with Visual Attention baseline for Chest X-Ray report generation and progressively investigate more advanced multimodal architectures involving abnormality detection, anatomical grounding, localization or segmentation, structured clinical representation, uncertainty estimation, evidence retrieval and multimodal reasoning.

Clinical reliability and safety are major design considerations. Therefore, the system will evaluate not only report-generation quality but also diagnostic performance, factual consistency, anatomical grounding, localization quality, uncertainty calibration, hallucination, omission, contradiction and failure cases. Low-confidence or insufficiently supported outputs will be explicitly flagged for human review.

MEDIVISTA AI is intended as an educational and academic research prototype. It is not a clinically certified medical device and is not designed to replace radiologists or other qualified healthcare professionals. The final clinical interpretation, diagnosis, treatment decision and prescription remain the responsibility of an appropriately qualified healthcare professional.

1.2 Research Vision

The long-term research vision of MEDIVISTA AI is to investigate how multimodal medical AI can support the complete medical imaging interpretation workflow across multiple imaging modalities while maintaining evidence traceability, uncertainty awareness and human oversight.

The intended research progression is:

Medical Image → Modality-Specific Visual Understanding → Finding Detection → Anatomical Localization / Segmentation → Measurements → Structured Clinical Representation → Multimodal Reasoning → Grounded Radiology Report → Evidence Retrieval → Clinical Decision Support → Human Review

The system will support four major imaging modalities:

X-Ray
CT
MRI
Ultrasound

Each modality may require a specialized image-processing and vision model because the characteristics of the data are different. X-Ray primarily involves 2D projection images, CT involves volumetric cross-sectional data, MRI provides multi-sequence anatomical information, and Ultrasound contains dynamic imaging information with modality-specific acquisition characteristics.

Rather than developing one monolithic model for every task, the research architecture will investigate a modular approach consisting of modality-specific vision components, a shared structured clinical representation, multimodal reasoning, evidence retrieval and safety mechanisms.

A central research objective is to establish an Evidence Chain connecting visual observations to generated clinical information:

Image Evidence → Finding → Anatomical Region → Measurement → Structured Finding → Report Statement → Supporting Evidence → Recommendation

This approach is intended to improve the traceability and factual consistency of generated reports and clinical decision-support outputs.

The research will also investigate longitudinal analysis, where previous and current imaging studies can be compared to identify changes, progression or stability in clinically relevant findings.

Clinical decision support will be designed as an evidence-grounded and human-reviewed capability. Retrieved medical evidence and structured imaging findings will be used to provide contextual diagnostic and management-oriented information. Treatment-related outputs may be presented as clinician-reviewable drafts rather than autonomous prescriptions.

The research will evaluate the system at multiple levels, including abnormality detection, localization or segmentation, measurement accuracy, report factuality, grounding quality, uncertainty calibration, evidence retrieval, hallucination, omission, contradiction and generalization across imaging modalities.

The overall vision is therefore not to replace radiologists, but to investigate an AI copilot that can connect medical images, structured findings, generated reports and supporting evidence into a traceable workflow that assists qualified healthcare professionals and researchers.

1.3 Project Scope

The system will support four medical imaging modalities: X-Ray, CT, MRI and Ultrasound.

The project will cover:

- Medical image upload and modality identification
- Abnormality and disease/finding detection
- Anatomical localization and segmentation where applicable
- Measurement extraction where applicable
- Structured clinical finding generation
- Evidence-grounded radiology report generation
- Confidence and uncertainty estimation
- Hallucination, omission and contradiction detection
- Evidence retrieval using RAG
- Clinical reasoning and decision support
- Longitudinal comparison of imaging studies
- Clinician-reviewable treatment/management suggestions
- AI Evidence Chain for traceability
- Model evaluation, failure analysis and ablation studies

The system will remain an academic research prototype and will not autonomously diagnose, prescribe medication, make final treatment decisions or replace qualified healthcare professionals.

1.4 Expected Outcome

The expected outcome is a research-oriented multimodal medical AI prototype capable of transforming a chest X-ray into a structured, grounded and uncertainty-aware clinical interpretation, while providing a foundation for future research into more comprehensive autonomous clinical decision support.

SECTION 02

Problem Statement

2.1 Problem Statement

Medical imaging interpretation is a complex process that requires identifying abnormalities, understanding anatomical context, assessing clinical significance and generating accurate reports.

Different modalities such as X-Ray, CT, MRI and Ultrasound provide different types of medical information, making automated interpretation challenging.

Existing AI systems often focus on a single task such as disease detection or image-to-text report generation. Such systems may generate fluent outputs while missing findings, producing unsupported statements or failing to provide sufficient evidence for their conclusions.

The core problem addressed by this project is:

How can a multimodal AI system analyze X-Ray, CT, MRI and Ultrasound images, identify clinically relevant findings, connect them to anatomical evidence, generate grounded reports and provide evidence-based clinical decision support while maintaining uncertainty and human oversight?

The proposed system addresses this through:

1. Modality-Specific Image Understanding
2. Abnormality and Disease/Finding Detection
3. Anatomical Localization / Segmentation
4. Measurement Extraction
5. Structured Clinical Representation
6. Grounded Report Generation
7. Evidence Retrieval and Clinical Reasoning
8. Clinical Decision Support
9. Uncertainty Estimation and Safety Checks
10. Human Review and Approval

2.2 Key Challenges

The project focuses on the following research challenges:

- Accurate detection of abnormalities from chest X-ray images.
- Generating clinically meaningful rather than merely linguistically similar reports.
- Preventing unsupported findings and hallucinations.
- Establishing a relationship between generated findings and image regions.
- Quantifying uncertainty in model predictions.
- Maintaining consistency between detected abnormalities and generated reports.
- Evaluating model performance using clinical as well as natural-language metrics.
- Investigating generalization across different imaging conditions and populations.
- Establishing a reliable foundation for future clinical decision-support capabilities.

2.3 Proposed Approach

The proposed system will use a multimodal AI architecture that separates chest X-ray interpretation into multiple interconnected stages rather than directly converting an image into free-form text.

The system will first extract visual representations from the chest X-ray and identify clinically relevant abnormalities. Detected findings will then be associated with relevant anatomical regions to provide visual grounding and improve the factuality of the generated interpretation.

The identified findings will be converted into a structured clinical representation and passed to a multimodal reasoning component for report generation. The system will generate a structured Findings and Impression report while estimating confidence and uncertainty for its predictions.

A further layer will provide evidence-based clinical decision support by using the structured findings and relevant clinical evidence to generate appropriate diagnostic, risk-assessment, and management-oriented recommendations.

The system will also incorporate mechanisms for detecting unsupported findings, omissions, contradictions, and low-confidence cases.

The proposed approach will be evaluated against a CNN-LSTM with visual attention baseline to determine whether the multimodal architecture improves clinical accuracy, report factuality, visual grounding, uncertainty calibration, and hallucination control.

Flow

Chest X-Ray → Visual Understanding → Abnormality Detection → Anatomical Grounding → Structured Findings → Multimodal Reasoning → Radiology Report → Clinical Decision Support

2.4 Expected Impact

The proposed system is expected to demonstrate the feasibility of using multimodal AI to automate multiple stages of the chest X-ray interpretation workflow. Instead of limiting AI to report drafting, the project aims to establish an integrated pipeline covering image understanding, abnormality detection, anatomical grounding, clinical reasoning, report generation, and evidence-based clinical decision support.

The research is expected to provide measurable improvements in the reliability and clinical relevance of automated radiology reporting by reducing unsupported findings, identifying uncertainty, and maintaining a clear relationship between detected abnormalities and generated report statements.

From a research perspective, the project will provide a framework for comparing conventional CNN-LSTM-based report generation with a more comprehensive multimodal approach. The

results, ablation studies, and failure analysis will be used to identify the strengths, limitations, and future potential of autonomous medical AI systems.

The long-term impact of the project is to investigate how AI can progressively automate radiology and clinical decision-support workflows while maintaining measurable standards of accuracy, reliability, transparency, and evidence-based reasoning.

SECTION 03

Business Requirements Document (BRD)

3.1 Business Problem

The increasing volume of chest X-ray examinations creates a significant demand for efficient, accurate, and scalable medical image interpretation. Traditional radiology workflows require trained professionals to manually examine images, identify abnormalities, assess their clinical significance, and prepare structured reports. This process can become time-intensive, particularly when large numbers of studies must be reviewed.

Existing AI-based systems often focus on individual tasks such as abnormality classification or automated report generation. A system that integrates visual understanding, abnormality detection, anatomical grounding, uncertainty estimation, clinical reasoning, and report generation can provide a more comprehensive approach to automated chest X-ray interpretation.

The business and research problem addressed by this project is therefore to develop an AI-based platform capable of performing multiple stages of the chest X-ray interpretation workflow through a unified multimodal architecture. The platform should generate clinically meaningful and visually grounded interpretations while identifying uncertainty and unsupported predictions.

The system will additionally establish a foundation for evidence-based clinical decision support, enabling future integration of patient-specific clinical information and medical guidelines. The objective is to investigate the extent to which such an integrated AI system can automate and improve the efficiency, consistency, and reliability of the radiology workflow.

3.2 Business Objectives

BO-01: Develop an end-to-end AI platform capable of processing chest X-ray images and producing structured clinical interpretations.

BO-02: Automate multiple stages of the radiology workflow, including abnormality detection, anatomical localization, report generation, and clinical decision support.

BO-03: Provide a scalable software architecture through a FastAPI backend and React-based interface for demonstrating the complete AI workflow.

BO-04: Provide transparent outputs including detected abnormalities, confidence estimates, visual grounding, generated reports, and supporting clinical evidence.

Research Objectives

RO-01: Develop and evaluate a multimodal vision-language architecture for automated chest X-ray interpretation.

RO-02: Compare the proposed architecture against a CNN-LSTM with visual attention baseline.

RO-03: Investigate whether structured abnormality detection and anatomical grounding improve the factuality and reliability of generated radiology reports.

RO-04: Develop an uncertainty-aware mechanism capable of identifying low-confidence predictions and unsupported clinical conclusions.

RO-05: Evaluate the system using diagnostic, report-generation, grounding, factuality, uncertainty, and safety-related metrics rather than relying only on text-overlap metrics.

RO-06: Investigate the feasibility of extending automated radiology interpretation toward evidence-based clinical decision support.

RO-07: Perform ablation studies and failure analysis to identify which components contribute most to the overall system performance.

3.3 Functional Requirements

The system shall provide the following functional capabilities:

FR-001 — Multimodal Input

The system shall accept X-Ray, CT, MRI and Ultrasound studies.

FR-002 — Image Quality Assessment

The system shall validate image quality and identify unsupported or inadequate inputs.

FR-003 — Finding Detection

The system shall detect relevant abnormalities or clinical findings with confidence scores.

FR-004 — Localization / Segmentation

The system shall localize or segment findings where appropriate for the imaging modality.

FR-005 — Structured Clinical Representation

The system shall convert detected findings into structured clinical information including anatomy, measurements and confidence.

FR-006 — Clinical Reasoning

The system shall perform multimodal reasoning using visual findings and available clinical context.

FR-007 — Grounded Report Generation

The system shall generate structured Findings and Impression sections supported by detected evidence.

FR-008 — Uncertainty Estimation

The system shall estimate prediction confidence and uncertainty.

FR-009 — Evidence Chain

The system shall link findings and report statements to relevant image evidence and supporting information.

FR-010 — Hallucination & Consistency Checking

The system shall identify unsupported, contradictory or omitted findings.

FR-011 — Evidence Retrieval

The system shall retrieve relevant medical evidence using an evidence-retrieval/RAG component.

FR-012 — Clinical Decision Support

The system shall provide evidence-based diagnostic and management-oriented decision support.

FR-013 — Longitudinal Comparison

The system shall support comparison of current and previous imaging studies where applicable.

FR-014 — Human Review

The system shall allow qualified users to review, edit and approve AI-generated reports and recommendations.

FR-015 — Evaluation Module

The system shall provide evaluation of diagnostic, report-generation, grounding, uncertainty and safety metrics.

FR-016 — Auditability

The system shall maintain model versions, inference results, confidence information, reports and relevant system events.

3.4 Success Criteria

The project will be considered successful if it demonstrates:

- Reliable finding/disease detection across supported modalities.
- Accurate localization, segmentation and measurements where applicable.
- Factual and clinically coherent Findings and Impression sections.
- Traceable Evidence Chain linking image evidence to report statements.
- Calibrated confidence and uncertainty estimation.
- Reduced hallucination, omission and contradiction in generated reports.
- Relevant evidence retrieval through the RAG layer.
- Useful clinical decision-support outputs with human review.
- Successful longitudinal comparison where prior studies are available.
- Measurable improvement over the initial Chest X-Ray CNN + LSTM baseline.
- Complete model evaluation, failure analysis and auditability.

3.5 Assumptions & Constraints

The following assumptions and constraints define the boundaries of the current research prototype:

- The system will initially use the **MIMIC-CXR dataset and its official train/validation/test split**.
- The dataset and its associated reports will be used only for research and model development within the permitted usage conditions.
- The initial system will focus on **chest X-ray images and English-language radiology reports**.
- The system will operate on already-acquired digital chest X-ray images and will not control the image acquisition process.
- The project will be developed using academic-scale computational resources, including a single-GPU workstation or cloud GPU where available.
- The proposed system is a research prototype and does not represent a clinically certified or regulator-approved medical device.
- Clinical decision-support outputs will depend on the quality and completeness of the available image evidence and supporting clinical information.
- Medication or treatment recommendations cannot be considered universally applicable without patient-specific information such as medical history, allergies, laboratory results, current medications, and other relevant clinical factors.
- The system shall provide uncertainty information and may abstain from generating conclusions when available evidence is insufficient.
- External clinical validation, prospective clinical trials, regulatory certification, and real-world hospital deployment are outside the scope of the current academic prototype.
- The dataset selection may be revised in consultation with the project mentor without changing the overall system architecture and research objectives.

3.6 Scope

The scope of the project is to develop and evaluate an end-to-end multimodal AI research prototype for chest X-ray interpretation. The system will extend beyond simple report generation by integrating visual understanding, abnormality detection, anatomical grounding, uncertainty estimation, clinical reasoning, and evidence-based clinical decision support.

In Scope

- Chest X-ray image upload and preprocessing.
- Image quality assessment and input validation.
- Automated detection of clinically relevant abnormalities.
- Confidence estimation for detected findings.
- Anatomical grounding of detected abnormalities.
- Structured representation of clinical findings.
- Multimodal clinical reasoning.
- Automated generation of Findings and Impression sections.
- Uncertainty estimation and low-confidence case identification.
- Hallucination, omission, and consistency analysis.
- Model abstention for insufficient-confidence cases.
- Evidence retrieval for clinical decision support.
- Evidence-based diagnostic and management-oriented recommendations.
- Comparison with a CNN-LSTM with visual attention baseline.
- Quantitative evaluation, ablation studies, and failure analysis.
- FastAPI-based inference and research APIs.
- React-based interface for demonstrating the complete workflow.
- Model versioning, experiment tracking, and evaluation reporting.

Out of Scope for the Current Prototype

- Regulatory approval or clinical certification.
- Prospective clinical trials or real-world clinical deployment.
- Direct autonomous prescribing of medication to real patients.
- Direct replacement of licensed healthcare professionals in clinical practice.
- Integration with production hospital PACS/EHR systems.
- Diagnosis based exclusively on X-ray when additional clinical information is required.
- Guaranteed treatment recommendations without sufficient patient-specific clinical information.
- Training on private hospital data that is not available under an appropriate research agreement.

The excluded capabilities represent potential future research directions rather than claims of capability for the current academic prototype.

3.7 Risks

Risk	Impact	Mitigation
Model hallucination	Incorrect findings or report statements	Grounding, consistency checking and uncertainty estimation
False negative findings	Important abnormality may be missed	Evaluate sensitivity/recall and critical finding performance
False positive findings	Incorrect abnormality detection	Confidence thresholds and validation
Dataset bias	Reduced generalization	Evaluate on appropriate validation/test data
Model overconfidence	Incorrect predictions presented with high confidence	Uncertainty estimation and abstention
Clinical recommendation error	Potentially inappropriate decision support	Evidence-based retrieval and clear uncertainty reporting
Limited computational resources	Slower training and experimentation	Transfer learning and optimized training

SECTION 04

RESEARCH QUESTIONS & HYPOTHESIS

4.1 Research Questions

RQ1: How effectively can AI detect clinically relevant findings across X-Ray, CT, MRI and Ultrasound?

RQ2: Can structured clinical representations improve the factuality and consistency of generated radiology reports?

RQ3: Does an Evidence Chain improve the traceability of AI-generated findings and report statements?

RQ4: Can RAG-based evidence retrieval improve clinical decision-support reliability?

RQ5: How effectively can uncertainty estimation identify low-confidence predictions and unsafe outputs?

RQ6: Can the proposed multimodal architecture generalize across different imaging modalities?

4.2 Research Hypothesis

A modular multimodal medical AI architecture combining modality-specific image understanding, structured clinical representation, evidence-grounded reasoning and uncertainty estimation will produce more factual, traceable and clinically useful outputs than a direct image-to-report generation approach.

The hypothesis will be evaluated through diagnostic, report-generation, grounding, evidence-retrieval and safety metrics.

SECTION 05

RESEARCH GAP & PROPOSED CONTRIBUTIONS

5.1 Research Gap

Existing medical AI research commonly focuses on individual tasks such as disease classification, image segmentation or report generation.

Key gaps include:

- Limited unified workflows across multiple imaging modalities.
- Weak connection between detected findings and generated report statements.
- Limited anatomical grounding and evidence traceability.
- Hallucination and omission in generated clinical text.
- Insufficient uncertainty and safety handling.
- Limited integration of evidence retrieval with clinical reasoning.
- Limited support for longitudinal imaging comparison.

MEDIVISTA AI addresses these gaps through a modular multimodal pipeline and an Evidence Chain connecting image observations to clinical outputs.

5.2 Proposed Contributions

The proposed research contributions are: 1. A multimodal architecture supporting X-Ray, CT, MRI and Ultrasound. 2. A structured clinical representation connecting visual findings with report generation. 3. An Evidence Chain for traceable AI reasoning. 4. Grounded radiology report generation with Findings and Impression sections. 5. RAG-based evidence retrieval for clinical decision support. 6. Uncertainty, hallucination and consistency checks. 7. Longitudinal comparison of imaging studies. 8. A research-oriented evaluation, failure-analysis and audit framework.

SECTION 06

END USERS & STAKEHOLDERS

6.1 End Users

| User | Requirement | |---|---| | Researchers | Train, evaluate and compare multimodal AI models | | Medical Professionals | Review AI findings, reports and decision-support outputs | | Clinical Administrators | Monitor system performance, studies and model versions | | System Administrators | Manage users, access, configuration and audit logs |

6.2 Stakeholders

- Project Mentor / Academic Evaluators
 - AI/ML Researchers
 - Radiologists and Medical Professionals
 - Healthcare Technology Teams
 - Clinical Administrators
 - System Administrators
-

SECTION 07

OBJECTIVES

7.1 Primary Objectives

1. Develop a multimodal AI system supporting X-Ray, CT, MRI and Ultrasound.
2. Detect clinically relevant abnormalities and findings.
3. Perform anatomical localization, segmentation and measurements where applicable.
4. Generate structured, grounded Findings and Impression sections.
5. Build an Evidence Chain linking findings to image and clinical evidence.
6. Estimate confidence and uncertainty.
7. Reduce hallucination, omission and contradiction.
8. Provide evidence-grounded clinical decision support.
9. Support longitudinal comparison of imaging studies.
10. Evaluate the system through diagnostic, report, grounding and safety metrics.

7.2 Secondary Objectives

- Develop a FastAPI-based AI inference backend.
- Develop a React-based medical imaging workstation.
- Implement RAG-based evidence retrieval.
- Maintain model versions and experiment results.

- Perform ablation studies and failure analysis.
 - Maintain audit logs and human-review workflows.
-

SECTION 08

KEY TERMINOLOGY

Multimodal AI — AI system supporting multiple medical imaging modalities.

Vision-Language Model — Model connecting visual information with language understanding and generation.

Abnormality Detection — Identification of clinically relevant findings in medical images.

Anatomical Grounding — Linking findings to relevant anatomical image regions.

Segmentation — Pixel-level or voxel-level identification of anatomical structures or abnormalities.

Clinical Reasoning — Processing imaging findings and clinical context to support diagnostic or management decisions.

Structured Clinical Representation — Standardized representation of findings, anatomy, measurements and confidence.

Evidence Chain — Traceable link from image evidence to finding, report statement, supporting evidence and recommendation.

RAG — Retrieval-Augmented Generation using external evidence to support generated outputs.

Uncertainty Estimation — Measurement of confidence and uncertainty in model predictions.

Hallucination — Generation of a clinical statement unsupported by available evidence.

Model Abstention — Avoiding a conclusion when confidence or evidence is insufficient.

Longitudinal Comparison — Comparison of current and previous imaging studies.

Clinical Decision Support — AI-generated information intended to support qualified clinical decision-making.

SECTION 09

DATA DESCRIPTION

9.1 Dataset Strategy

The project will use modality-specific datasets for X-Ray, CT, MRI and Ultrasound.

The initial research and baseline pipeline will use MIMIC-CXR with its official train/validation/test split. It provides de-identified chest X-ray images paired with radiology reports. :contentReference[oaicite:0]{index=0}

9.2 Modality Data

Modality	Data Type	Primary Tasks
X-Ray	2D images + reports	Detection, grounding, report generation
CT	2D slices / 3D volumes	Detection, segmentation, measurement
MRI	Multi-sequence images	Detection, segmentation, comparison
Ultrasound	2D / dynamic images	Finding detection and analysis

9.3 Data Requirements

Datasets should preferably contain:

- De-identified medical images
- Appropriate disease/finding labels
- Anatomical annotations where available
- Segmentation/localization annotations where available
- Reports or clinical descriptions where available
- Valid train/validation/test splits

9.4 Initial Research Dataset

MIMIC-CXR will be used as the primary dataset for developing and evaluating the initial Chest X-Ray pipeline and CNN-LSTM baseline.

SECTION 10

SYSTEM OVERVIEW

10.1 System Overview

MEDIVISTA AI is a multimodal medical imaging AI platform supporting X-Ray, CT, MRI and Ultrasound.

The overall workflow is:

Medical Image → Modality Identification → Image Quality Check → Modality-Specific AI Analysis → Finding Detection → Localization / Segmentation → Measurements → Structured

Clinical Representation → Multimodal Reasoning → Grounded Report Generation → Evidence Retrieval → Clinical Decision Support → Human Review

The system uses modality-specific vision models rather than a single model for all imaging types.

A shared structured representation connects visual findings with report generation, evidence retrieval and clinical reasoning.

The core Evidence Chain is:

Image Evidence → Finding → Anatomy → Measurement → Report Statement → Supporting Evidence → Recommendation

10.2 User Flow

Login → Create Study → Enter Patient/Study Information → Upload Imaging Study → Select/Detect Modality → AI Analysis → Review Findings → Inspect Evidence/Confidence → Generate Report → Clinical Copilot → Longitudinal Comparison (if available) → Review & Edit → Approve/Export

10.3 Research / Model Flow

Image → Modality-Specific Encoder → Detection / Segmentation → Anatomical Grounding → Measurements → Structured Clinical Representation → Multimodal Reasoning → Report Generation → RAG Evidence Retrieval → Clinical Decision Support → Uncertainty & Safety Checks → Evaluation

SECTION 11

UX Requirements

11.1 UX Objective

The MEDIVISTA AI interface will provide a professional medical imaging workstation for researchers and qualified medical professionals.

The interface should make AI analysis transparent through:

- Imaging viewer with overlays and annotations
- Findings and confidence information
- Evidence Chain
- Grounded report editor
- Clinical reasoning and evidence

-
- Longitudinal comparison
 - Model evaluation and failure analysis
 - Human review and approval

The design should prioritize clarity, traceability, accessibility and safe human oversight.

11.2 Information Architecture

```
| Route | Purpose |  
|---|---|  
| /overview | System overview and recent studies |  
| /new-study | Create and upload a new study |  
| /studies | Browse imaging studies |  
| /studies/{id} | Imaging viewer and AI analysis |  
| /reports/{id} | Grounded report editor |  
| /copilot | Clinical reasoning and decision support |  
| /comparison | Longitudinal study comparison |  
| /evidence | Evidence Library and RAG |  
| /evaluation | Model performance and metrics |  
| /experiments | Research experiments |  
| /models | Model Registry |  
| /failures | Failure Analysis |  
| /audit | Audit logs |
```

11.3 Screen Requirements

CLINICAL WORKSPACE

Dashboard

New Analysis

Studies

AI Analysis & Viewer

Anatomical Grounding

Generated Reports

Clinical Support

Model Evaluation

STUDIES / CXR-2024-8831-PA / Spatial Anatomical Grounding & Cross-Attention

ENCODER-DECODER ATTENTION LAYER L24-H8

Dice Similarity Index: 0.892 | Spatial Latency: 18.4ms

Labeling Visualizations: AI Grounding & Spatial Attention
Heatmaps — Indicative neural focus, not definitive histological pathology.

CROSS-ATTENTION

PROPOSAL BOXES (IOU > 0.5)

CONTOURS

GRAD-CAM++

OVERLAY OPACITY

70%

THRESHOLD CUTOFF

0.75

ANATOMICAL LANDMARKS:

VISIBLE (4 ANCHORS)

RESET SALIENCY

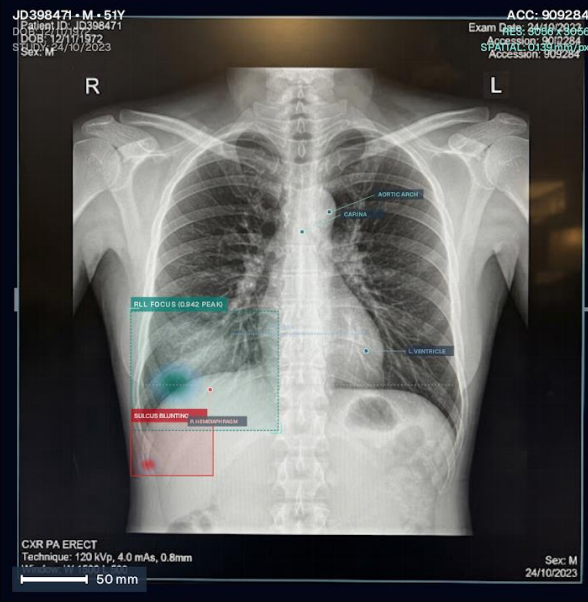
EXPORT SALIENCY MAP

VIEWPORT A // PA
ERECT

120 kVp • 4.0 mAs

Window W: 1500 L: 500

Zoom: 100% (1:1 Native)



SPLIT COMPARE WITH BASELINE PA

GRAYSCALE / PSEUDO-THERMAL

Cross-Entropy Loss: 0.041

SAM-Med3D Grounded

GROUNDING REGIONS & ATTENTION DENSITY

3 DETECTIONS
ACTIVE

Region 01: RLL
Consolidation Focus

93.8% GROUNDING
CONF.

X COORD	Y COORD	BOUNDING (WXH)	PIXEL PEAK
584 px	612 px	210 x 185	0.942

Dense alveolar opacity effacing the medial right hemidiaphragmatic contour, air bronchograms visualized within segment 8/9.

ASSOCIATED NLP SALIENCY TOKENS:

consolidation (0.91) opacity (0.87)

air bronchogram (0.78) right lower lobe

Region 02: Right
Costophrenic Sulcus

81.6% GROUNDING
CONF.

BBox [X: 690, Y: 740, W: 110, H: 90]

Attribution Peak:
0.816

Meniscus-like fluid density with 8mm blunting of the posterior/lateral costophrenic recess consistent with reactive pleural effusion.

Region 03: Cardiac
Silhouette & CTR

CTR: 0.44
(NORMATIVE)

Transverse Cardiac Diameter: 13.2 cm

Ref: < 0.50 (Normal)

CLINICIAN GROUNDING
FEEDBACK

Audit Stamp: VANCE-M-882

Validate spatial grounding accuracy for the selected anatomical proposal (Region 01: RLL). Feedback tunes continuous federated gradients.



ACCURATE
ANATOMY



MISALIGNED /
ARTIFACT



OVERSIZED REGION

SYNC WITH REPORT GENERATOR

Settings

INFERENCE ENGINE

Ready

MedVision AI

DIAGNOSTICS V2.4

CLINICAL WORKSPACE

Dashboard

New Analysis

Studies

AI Analysis & Viewer

Anatomical Grounding

Generated Reports

Clinical Support

Model Evaluation

Settings

INFERENCE ENGINE

Ready

V2.4-RC | MIMIC-CXR Fine-Tuned

Search Patient ID, DICOM Accession, or

PACS Online • Latency 14ms

Dr. Marcus Vance, MD, PhD

Lead Clinical AI Investigator

STUDIES / AUDIT DOSSIER / CXR-2024-8831-PA

Study Archive: CXR-2024-8831-PA

FDA 510(K) CLEARED V2.4

DICOM SOP verified

EXPORT PACS BUNDLE

SIGNED CLINICAL STAMP

PATIENT HASH ID

MODALITY & PROJECTION

ACQUISITION TIMESTAMP

INFERENCE LATENCY

CONFIDENCE CALIBRATION

AUDIT CONSENSUS

#SUB-77402

CXR PA ERECT (120 kVp)

Oct 24, 2023 08:42:15 UTC

1.42s (MIMIC-v4 Core)

Platt-Calibrated 99.1%

Physician Concordant

UNIFIED SUMMARY (ACTIVE)

DICOM IMAGE & OVERLAYS

GROUNDING ATTENTION MAPS

GENERATED RADIOLOGY REPORT

CLINICAL DEC

DICOM Viewport (0020,0032)

NATIVE 2484X3024

PATIENT: JD398471

DOB: 12/11/1972 (51Y)

SEX: MALE | LAT: R

ACCESSION: #909284

EXAM: OCT 24, 2023

KV: 120 | MAS: 4.0

THICKNESS: 0.8mm

WINDOW: W1500 L500

RIGHT LOWER LOBE (RLL) CONSOLIDATION (94.2%)

COSTOPHRENIC BLUNTING (82.5%)

ROI #01 [DICE: 0.892]

MASK: CAM + Seg

PULMONARY LUNG

MEDIASTINUM

BONE WINDOW

DICOM STANDARD METADATA

ISO 12052

Pixel Spacing: 0.139 / 0.139 mm

Photometric: MONOCHROME2

Detector: Flat Panel Amorphous

Exposure Index: 312 [Target 300]

Panel 1: Detected Findings & Calibrated Confidence

CRITICAL REVIEW

Right Lower Lobe (RLL) Consolidation

94.2% CONFIDENCE

Marked alveolar opacity obscuring the medial right hemidiaphragmatic contour, compatible with lobar pneumonia. No signs of volumetric collapse.

Right Costophrenic Sulcus Blunting

82.5% CONFIDENCE

Small-to-moderate reactive parapneumonic pleural effusion noted along the lateral right pleural gutter. Left pleural space remains clear.

Bilateral Hilar Prominence

AI ABSTAINED (CONFORMAL THRESHOLD < 0.80)

Safety protocol engaged: Ambiguous pulmonary vasculature overlap vs. hilar lymphadenopathy. Algorithm deferral to manual fellowship-trained overread.

Panel 2: Spatial Grounding Geometry & Telemetry

Intersection Over Union: 0.814

ROI TARGET

BOUNDING POLYGON (X, Y, W, H)

ANATOMICAL ANCHOR

DICE COEFF

VOLUME EST.

#01 RLL Consolidation

[584, 612, 210, 185]

Right Inferior Lobe (Segment 8/9)

0.892

184.2 cm³

#02 Effusion Sulcus

[692, 840, 142, 94]

Right Posterior/Lateral Recess

0.865

48.5 cm³

#03 Heart / CTR

[340, 510, 380, 240]

Cardiothoracic Ratio 0.46 (Normal)

0.954

Normative

Panel 3: Automated Radiology Report & Attestation

CLINICIAN SIGNED

STRUCTURED IMPRESSION (RADLEX ANNOTATED)

IMPRESSION: Acute dense consolidation in the right lower lobe consistent with community-acquired or aspiration bacterial pneumonia. Accompanied by minimal right reactive pleural effusion. No evidence of pneumothorax. Cardiothoracic silhouette is within normal limits.

Digitally Validated: Marcus Vance, MD, PhD, DABR

NPI: #1892841029 • Clinical Associate Professor of Radiology

TIMESTAMP: 2023-10-24T08:58:02Z

ED FAST-TRACK TIER 1 NOTIFICATION SENT

Panel 4: Clinical Decision Support & Guideline Pathways

ATS / IDSA GUIDELINES 2023

IMMEDIATE STEP 01

Microbiology Workup

Order 2 sets of peripheral blood cultures and rapid sputum Gram stain prior to broad-spectrum antibiotic initiation.

Order Ready in EHR

IMMEDIATE STEP 02

CURB-65 Triage Stratification

Age 51 (Score 0). Assess outpatient vs observational admission based on RR ≥ 30 and BUN levels.

Score 1 (Low-Intermediate)

FOLLOW-UP STEP 03

Surveillance Radiography

Schedule follow-up PA and lateral chest radiograph at 6 to 8 weeks to document radiographic resolution and exclude endobronchial malignancy.

Dec 12, 2023 Target

Panel 5: AI Model Lineage & Algorithmic Provenance

SEC-AUDIT-PASS

MODEL CHECKPOINT SHA-256

DICOM SOP INSTANCE UID

IRB & ETHICAL PROTOCOL CERTIFICATION

HARBOR COMPLIANT

1.2.840.113619.2.55.3.60468831.84920.1698136935

1.2.840.113619.2.55.3.60468831.84920.1698136935

HARDWARE CLUSTER TELEMETRY

Node gpu-c4-a100-node09 • FP16 TensorRT Engine

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V2.4-RC | MIMIC-CXR Fine-Tuned

Search Patient ID, DICOM Accession, or S

🔍

PACS Online • Latency 14ms

Dr. Marcus Vance, MD, PhD

Lead Clinical AI Investigator

Clinical Decision Support Engine • MULTI-MODAL EVIDENCE GROUNDING

Subject ID: SUB-77402 (Male, 51y) • DICOM Accession: 909284 • Presentation: Acute Respiratory ...

EXAM TIMESTAMP

24/10/2023 14:12 EST

MODALITY / VIEW

CXR PA Erect (120 kVp)

MODEL BACKBONE

BioViL-T Multi-Scale VIT

INFERENCE LATENCY

184 ms (Calibrated)

CURB-65 ESTIMATE

Score 1 (Low/Intermed)

VALIDATION STATE

Concordant

CXR GROUNDING PORT

PA 0.8mm SLICE

W: 1500 L: 500

ROI: R-BASILAR

Scale: 0.139 mm/px

Matrix: 2048 x 2048

Calibrated ROI

FITBIT_HOURLY_ACTIVITY

KEY A BIOM/

Right Lower Lobe Consolidation

ACUTE FOCUS

Statistical Confidence

94.2%

Severity Stratification

Moderate-Severe

Homogeneous alveolar opacity with distinct air bronchograms localized to the right posterior basal pulmonary segment.

Reactive Pleural Effusion

SECONDARY

Statistical Confidence

82.5%

Severity Stratification

Mild (Blunting)

Blunting of the right lateral costophrenic angle; meniscus visible without significant mediastinal shift.

PERTINENT NEGATIVES (RULING OUT EMERGENCIES)

☑️ Tension Pneumothorax

Negative (99.1% conf)

☑️ Cardiomegaly / Cardiac Index

CTR ≤ 0.48 (97.8% conf)

QUANTITATIVE RADIOMIC METRICS

Lung Opacity Volume Fraction

14.2% of right hemithorax

Consolidation Density Estimate

-45 HU Eq (Solid Exudate)

DIFFERENTIAL DIAGNOSTIC CONSIDERATIONS

PROBABILISTIC BAYESIAN PRIOR

PRIMARY PRIOR

ICD-10 J18.1

Community-Acquired Pneumonia (CAP)

POSTERIOR P(D)

89%

Acute Lobar Bacterial Presentation (e.g. S. pneumoniae, H. influenzae)

RADIOLOGICAL GROUNDING

Unilateral focal dense alveolar filling with patent air bronchograms limited to anatomical segments 9/10. Silhouette sign positive against right hemidiaphragm.

GUIDELINE CONCORDANCE

ATS / IDSA 2019 Consensus Guidelines

Criteria met for moderate-severity CAP. Outpatient vs observation bed triage subject to vital stability.

CURB-65 ESTIMATE

Score 1

Low-to-Intermediate 30d risk (2.7%)

PSI / PORT CLASS

Class II

Estimated points: 61

SECONDARY PRIOR

ICD-10 J69.0

Aspiration Pneumonitis / Anaerobic Infiltrate

POSTERIOR P(D)

12%

Distinguishing Factors: Dependent superior/basal segmental involvement in the right lower lobe commonly correlates with micro-aspiration during recumbency. Correlate with clinical swallow screening, neurological history, or altered consciousness episodes.

LONGITUDINAL CAUTION

Rule-Out

UNCERTAIN / BASELINE

Post-Obstructive Atelectasis / Endobronchial Lesion

Secondary to complete alveolar volume consolidation, endobronchial patency cannot be verified with plain radiography. Must schedule post-resolution follow-up in 6–8 weeks to assure complete clearance without persistent architectural distortion.

RISK & ESCALATION PREDICTORS

30-DAY MORTALITY

< 2.5%

Low Stratum

ICU ESCALATION RISK

4.8%

Standard Floor

COMPLICATED EFFUSION

14.1%

Monitor Closely

RECOMMENDED NEXT CLINICAL STEPS

ATS Tier 1

1 Immediate Diagnostic Workup

- Microbiology:** Collect deep expectorated Sputum Gram Stain & culture, and 2x blood cultures prior to initiation of targeted antimicrobial therapy.
- Biomarker Panel:** Serum Procalcitonin (PCT) and high-sensitivity CRP to assist clinical response monitoring.
- Urine Antigen Testing:** Leg. pneumophila and S. pneumoniae urinary assays if severe features emerge.

2 Radiographic Follow-up

- Scheduled PA Chest Re-check:** Repeat 2-view chest radiograph in **6 to 8 weeks** to verify complete parenchymal resolution and exclude hidden mass or endobronchial lesion.
- Radiographic lag behind clinical recovery is typical (up to 4–6 weeks for complete clearing).

3 Escalation Triggers

Prompt Thoracic Ultrasound or Contrast Chest CT is indicated if:

- Patient fails clinical temperature / PaO2 improvement within 48–72h.
- Parapneumonic effusion thickens or expands beyond 10mm rim.

APPEND RECOMMENDATIONS TO EHR ORDER

FLAG FOR PULMONARY CONSULTATION

MULTI-MODAL CONTEXT PIPELINE

FHIR R4 CONNECTED

EHR CLINICAL NOTES & HPI

14 Records Synced

LAB PANELS (CBC / CMP / ABG)

Pending Results (3)

ACTIVE PHARMACOTHERAPY

No Contraindications

CONTINUOUS VITALS STREAM

SpO2: 93% on Room Air

CLINICAL DECISION SUPPORT ADVISORY (INVESTIGATIONAL DECISION AID)

🛡️

Recommendations and probability assessments generated by the BioViL-T Clinical Reasoning Engine v2.4 are strictly calibrated decision aids for medical investigational evaluation. This system does NOT write prescriptions, schedule mandatory procedures, or establish standalone definitive diagnoses. All final clinical diagnostic decisions and therapeutic plans remain the sole ethical and legal responsibility of the licensed attending physician.

Model ID: CXR-DECISION-G3-70B • ISO 13485 / SaMD Class IIb

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Ready

V2.4-RC | MIMIC-CXR Fine-Tuned

Search Patient ID, DICOM Accession, or

PACS Online • Latency 14ms

Dr. Marcus Vance, MD, PhD

Lead Clinical AI Investigator

Study CXR-2024-8831-PA

Subject: SUB-77402 (JD398471) • Modality: Digital Radiography (Chest PA Erect)

AI DRAFT GENERATED • BioVIL-T Multimodal Synthesis • Pending Board Radiologist Attestation

REGENERATE DRAFT

DIFF PRIOR (14M)

EXPORT PACS PDF

SIGN & FINALIZE REPORT

SPATIAL AI EVIDENCE VIEWER

PA VIEW [2048x2048]

PT ID: JD398471

DOB: 12/11/1972 (51Y)

SEX: M

ACCESSION: 909284-01

DATE: 24/10/2023

TECH: 120 kVp, 4.0 mAs

R

L

ROI-01: RLL CONSOLIDATION (94.2%)

ROI-02: CPA BLUNTING (82.5%)

WINDOW: W:1500 L:500

SCALE: 0.139 mm/px

Click ROIs to align corresponding findings. Grad-CAM++ / Spatial Saliency

STRUCTURED EVIDENCE MATRIX

Calibration: $p < 0.001$

FINDING / REGION	SALIENCE	CONFIDENCE	DR. ACT
Right Lower Lobe Infiltrate/Consolidation	HIGH	94.2%	SYNTHESI.
Right Pleural Effusion (Trace)	MODERATE	82.5%	SYNTHESI.
Cardiomegaly / CTR > 0.50	NONE	97.8% Norm	EXCLUDI.
Pneumothorax (Bilateral)	NONE	99.1% Norm	EXCLUDI.

EPISTEMIC UNCERTAINTY GATE

ENTROPY = 0.54

Abstention Rule Activated: Right hilar prominence demonstrated epistemic variance exceeding algorithmic safety ceiling ($\tau > 0.40$). Omitted automatically from the definitive radiology draft to prevent false-positive lymphadenopathy citation. Manual radiologist inspection advised.

Ensemble Disagreement: 3/5 Models

AUDIT MODEL DISTRIBUTION →

DEPARTMENT OF RADIOLOGY & COMPUTATIONAL DIAGNOSTICS

Formal Diagnostic Radiology Report

CLINICAL AI-ASSISTED REPORTING SUITE • REGULATORY CLASS IIa WORKFLOW

UNVERIFIED PRELIMINARY DRAFT

Generated: 2023-10-24 14:32:08 UTC

ACCESSION#	ORDERING MD	SERVICE LOC	CLINICAL INDICATION
#909284-01	Dr. K. Evans, MD	Emergency Dept (ED-3)	Cough, Hypoxemia, ...

1. EXAMINATION

Chest Radiograph, Single View (PA Erect), DICOM 3.0 Standard Protocol

2. TECHNIQUE & IMAGE QUALITY

Adequate inspiratory effort demonstrated with visualization of 9 posterior ribs. Symmetric patient rotation without thoracic tilt. Exposure technique optimal at 120 kVp, 4.0 mAs, yielding standard diagnostic contrast resolution across pulmonary parenchyma and retrocardiac structures.

3. COMPARISON

Compared with baseline chest radiograph dated 14 months prior (12/08/2022). Previous study was normative with no parenchymal opacities or pleural effusions.

4. FINDINGS

AI GENERATED HIGHLIGHT

TOGGLE AI CLUES

MACROS

LUNGS: Dense airspace consolidation is identified in the right lower lobe with distinct air bronchograms, obscuring the right hemidiaphragmatic contour. Left lung fields are clear without focal infiltrates, nodular consolidation, vascular congestion, or pneumothorax.

PLEURA: Mild blunting of the right costophrenic angle consistent with small reactive pleural effusion. Left costophrenic angle is sharp. No signs of tension or hydropneumothorax.

CARDIOVASCULAR: Cardiac silhouette is normal in size with cardiothoracic ratio measured at 0.44. Mediastinal contours, aortic arch caliber, and pulmonary vascular distribution are unremarkable for age.

OSSEOUS / SOFT TISSUES: Thoracic osseous cage intact without acute rib fracture, cortical disruption, or aggressive osteolytic lesions. Chest wall soft tissues within normal limits.

5. IMPRESSION & CLINICAL SUMMARY

URGENT TIER 2

1. Dense right lower lobe airspace consolidation, highly concerning for acute community-acquired lobar bacterial pneumonia (Streptococcus pneumoniae or atypical respiratory pathogens).

2. Trace right-sided pleural effusion, likely reactive / uncomplicated parapneumonic effusion.

3. No evidence of pneumothorax, pulmonary edema, or cardiomegaly.

Recommendation: Clinical correlation, sputum culture, and follow-up CXR in 6 weeks to confirm resolution.

ACR Code: 3.210

EDIT FINDINGS DIRECTLY

RE-SYNTHESIZE WITH BIOVIL-T

Audit Trail Hash: SHA256-8c47f9...

ATTENDING RADIOLOGIST ATTESTATION

REQUIRED BY CFR TITLE 21 / JOINT COMMISSION

"I have personally reviewed the digital radiographs and clinical indications, scrutinized the computational segmentation overlays, and amended or verified the computer-assisted draft report findings above as accurate."

Signer: Marcus Vance, MD, PhD (Lic #RAD-991204)

UTC 2023-10-24 14:41:19

Type /Marcus Vance, MD/'

ATTEST & DISPATCH TO EHR

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+

MedVision AI

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INFERENCE ENGINE

Ready

V2.4-RC | MIMIC-CXR Fine-Tuned

Search Patient ID, DICOM Accession, or

PACS Online · Latency 14ms

Dr. Marcus Vance, MD, PhD
Lead Clinical AI Investigator

RESEARCH MULTI-SITE · DICOM Nodes Sync: 3 Online

Cohort: Stanford MIMIC-IV-CXR (n=14,892)

Export Telemetry CSV

Radiology AI Research Overview

Cohort Multi-Center Validation Study · Active Inference Engine: DenseNet-121 + BioClinical-BERT v2.4

Upload New Study

STUDIES ANALYZED

14,892

↑ 12.4%

PA/AP Cycle Run +1,640 this week

ABNORMALITIES DETECTED

3,418

22.9% POSITIVITY

Consolidation: 1,024 Effusion: 912 Cardiomegaly: 840

REPORTS GENERATED

12,904

96.2% Sign-off

Clinician Minor Rewrites 3.8% (490)

LOW-CONFIDENCE CASES

142

0.95% ABSTAINED

Review Consensus Req. 19 Pending Today

Recent Clinical Studies

Live Stream

Filter Density

STUDY ID / ACCESSION	TIMESTAMP	VIEW	PRIMARY AI FINDING & PROBABILITY	ENTROPY (H)	REVIEW STATUS	ACTION
CXR-89241-PA ACC: 909284 / JD398471	24/10/2023 14:18	PA	RLL Consolidation (94%)	0.12	Validated	INSPECT
CXR-89240-AP ACC: 909121 / PT084210	24/10/2023 14:02	AP	Bilateral Infiltrate (88%)	0.24	Pending Consensus	INSPECT
CXR-89239-PA ACC: 908994 / KL744192	24/10/2023 13:45	PA	Cardiomegaly (81%)	0.31	Validated	INSPECT
CXR-89238-LAT ACC: 908752 / RT119283	24/10/2023 13:22	LAT	Uncertain Hilar Density (AI Abstained 51%)	0.68	Flagged for Review	CONSENSUS
CXR-89237-PA ACC: 908641 / AA491024	24/10/2023 13:10	PA	Normal / No Abnormality (99%)	0.04	Validated	INSPECT

Showing 5 of 14,892 verified studies

Prev 1 / 2,979 Next

Model Benchmark Summary

Eval v2.4

AUROC (CheXpert 14-Labels)	0.932
Micro F1-Score	0.884
Clinical Factuality (RadGraph F1)	0.841
Anatomical Grounding mIoU	0.768

HALLUCINATION RATE
1.2%
Target < 2.0%

BRIER RELIABILITY
0.041
Strict Calibration

IRB GOVERNANCE

IEEE Clinical AI Compliance: System operates under Institutional Review Board Protocol #IRB-2023-884. Investigational platform; Research Use Only.

INFERENCE NODE TELEMETRY

18ms Q-Depth

VRAM Allocation
18.4 / 24.0 GB (A100)

Batch Throughput
32 imgs / sec

Subject CXR-89241-PA Telemetry

Active Selection

Anchor

Right lower lobe opacity compatible with focal bacterial consolidation. No visible pneumothorax or blunting of the costophrenic angles. Cardiothoracic ratio within physiological limits (0.44).

RadGraph F1
0.912

Anat. IoU
0.834

CheXbert Agree
Concordant

Audit Log

Open in AI Viewer

PA ERECT

W:1500 L:500

+

MedVision AI

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🔍

PACS Online · Latency 14ms

🔔

Dr. Marcus Vance, MD, PhD

Lead Clinical AI Investigator

MICCAI / IEEE BENCHMARK SUITE

RUN_ID: EVAL-2024-CXR-8819

Model Evaluation & Scientific Benchmark Suite

Formal multi-metric statistical verification for multimodal radiograph generative diagnostics.

LaTeX Table

W&B Artifact

Download IEEE Test Log

CANDIDATE MODEL (PROPOSED)

REFERENCE BASELINE

EVALUATION COHORT

CONSENSUS STANDARD

MedVision-DenseNet-BioViL v2.4

CheXNet-ResNet50 v1.2

MIMIC-CXR-JPG (n=3,269 test)

Triple-Board Certified (95% CI)

MACRO AUROC

MACRO AUPRC

MICRO F1 SCORE

RADGRAPH FACTUALITY

GROUNDING MIOU

CALIBRATION ECE

0.932

+4.8% vs 0.884

0.867

+7.6% vs 0.791

0.884

P: 0.892 · R: 0.876

0.841

Entity & Relation F1

0.768

Grad-CAM Heatmap Concordance

0.024

T-Scaled (T=1.14)

Comparative Model Performance (14 Pathologies)

BOOTSTRAP REPLICATES: B=1,000

PATHOLOGY ABNORMALITY	MEDVISION V2.4 (AUROC)	BASELINE (AUROC)	F1 SCORE	HALLUCINATION %	ABSTENTION %
Atelectasis	0.914	0.862	0.850	1.1%	4.2%
Cardiomegaly	0.958	0.920	0.930	0.4%	1.8%
Consolidation	0.946	0.891	0.910	1.2%	3.5%
Edema (Pulmonary)	0.938	0.895	0.890	0.8%	2.9%
Pleural Effusion	0.962	0.931	0.940	0.5%	1.2%
Pneumothorax (Tension/Simple)	0.971	0.918	0.950	0.3%	0.9%
Pneumonia	0.927	0.873	0.880	1.4%	3.1%

Uncertainty Abstention Protocol: 142 ambiguous examinations (0.95%) triaged to human radiologist oversight.

FALSE POSITIVE PENALTY: Reduced by 41.2%

Visual Grounding & Spatial Calibration Sample

DICOM UID: 1.2.840.113619.2.55.3.269

100mm

kVp: 120

mAs: 4.0

WL: -530 / WW: 1600

R-Bilateral Effusion IoU: 0.892

SPATIAL IOU VERIFICATION

DenseNet-BioViL incorporates cross-attention between token embeddings and high-resolution convolutional feature maps, achieving statistically significant alignment over pure bounding heuristics.

Bounding Box IoU (vs 3 Radiologists)

0.781 ± 0.04

Anatomical Precision (Point-pointing)

92.4%

P-value < 0.001 against CheXNet ResNet50 baseline

NLG Clinical Report Generation

BIOVIL-NLP

BLEU-4 SCORE

0.384

+0.093

Baseline: 0.291

ROUGE-L SCORE

0.428

+0.066

Baseline: 0.362

BERTSCORE (F1)

0.894

Semantic Sim

BioClinicalBERT Vector

HALLUCINATION RATE

1.2%

-7.5% Drop

Was 8.7% in v1.2

RADGRAPH FACTUALITY AGREEMENT

84.1% Exact F1

Evaluates structured entities: Anatomical Structure, Observation, and Modality Qualifier. Prevents clinically catastrophic negated findings.

Reliability & Probability Calibration

ECE: 0.024

Observed Accuracy vs Forecast Confidence

Temperature T = 1.14

Perfect Calibration

MedVision Post-T-Scaling

Observed Accuracy

Brier Score Loss:

0.071 (Sub-0.1 is optimal)

Negative Log Likelihood (NLL):

0.184

Under-Confidence Frequency:

< 0.4% total test set

CITATION & ALGORITHMIC PROVENANCE

MedVision AI Research Team, "Dense Multimodal Alignment and Calibrated Confidence Estimation for Automated Chest Radiography Triage." IEEE Transactions on Medical Imaging (TMI), 2024.

Copy BibTeX

Open Full Research PDF

MedVision AI

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PACS Online · Latency 14ms

Dr. Marcus Vance, MD, PhD

Lead Clinical AI Investigator

WORKFLOW MODULE / PIPELINE INGESTION

DICOM 3.0 REALTIME

ENGINE STATUS

BioViL-T / Active

QUEUE LATENCY

0.42 ms / Frame

COMPLIANCE MODE

HIPAA Safe Harbor

Initiate Chest Radiograph Analysis

Upload digital DICOM, PNG, or high-fidelity TIFF/JPEG for autonomous multi-label classification and report drafting.

Drag and drop DICOM (.dcm) or radiograph file here

or [browse local PACS archive](#) / network storage

DICOM 3.0 Part 1016-bit Grayscale TIFFLossless PNG / JPEG-LS

Max payload: 512MB

ACTIVE INGESTION NODE:

RAD-STORAGE-EAST-01 (C-STORE SCP)

Checksum Auto-Verify: SHA-256

DICOM INGESTION & QA VALIDATION PIPELINE

SERIES #0004 // 14-BIT MONOCHROME2

STUDY: CHEST_PA_ADULT

LAT: ERECT POSTEROANTER

DIM: 3056 x 3056 px (0.139mm)

R / PA

kVp: 120 | mA: 320

EXP: 4.0 mAs

WW: 1800

WL: -530 (Lung Window)

Lossless 100%

ALGORITHMIC QA GATES5 / 5 PASSED (100%)

Projection Validation

Detected: PA Erect

Confidence: 99.4%

PASS

Exposure Quality

Optimal Penetration

mAs: 4.0 | kVp: 120

PASS

Lung Field Completeness

Both Costophrenic Angles & Apices Fully Visualized

PASS

Motion Artifacts

Negligible / Sub-pixel stability Score: 0.02 MSE

PASS

Rotation Status

Symmetric Clavicular Alignment

0.4° tilt (Nominal)

PASS

EXTRACTED DICOM TAG

VIEW RAW HEADERS

REGISTRY

SOP UID: .12.840.113619

Photometric: MONO2

Bit Depth: 14 allocated

Detector: Flat Panel a-Si

RESEARCH STUDY METADATA

HIPAA DE-IDENTIFIED

RESEARCH SUBJECT ID

Safe Harbor Pseudonym

SUB-2024-CXR-9912

COHORT / STUDY ARM

Inpatient Critical Care - Cohort B (Pulmonary Infiltrates)

AGE BRACKET

BIOLOGICAL SEX

50-59 years

Bin #6

Female (XX)

CLINICAL INDICATION / PRESENTATION

Shortness of breath, low-grade fever, productive cough (3 days duration). Suspected community-acquired consolidation vs

ACQUISITION DEVICE / DETECTOR RIG

Siemens Healthineers Ysio MAX

SN: #SH-4091A

INFERENCE & SAFETY THRESHOLDS

CALIBRATED V2.4

CONFIDENCE THRESHOLD (T)

80% (STRICT CLINICAL SENSITIVITY)

50% (High Recall / Screening)80% (Strict Balanced)95% (High Specificity)

Uncertainty Abstinence Policy

ACTIVE

Force AI Abstinence when epistemic entropy H(y) > 0.45

GROUNDING SALIENCY ARCHITECTURE

Bilinear Bounding Mask

Grad-CAM++ with BioViL-T Cross-Attention

Res: 14x14 grid

RUN AUTONOMOUS AI INTERPRETATION

(EST. 1.2S)

RESET / UPLOAD ANOTHER

LOAD DEMO (PNEUMOTHORAX)

INVESTIGATIONAL DEVICE NOTICE // SAFE HARBOR ENFORCED

Research Use Only (RUO). MedVision AI is an investigational software suite evaluated under IRB oversight. Not for standalone diagnostic decision making without licensed board-certified radiologist verification.

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State	User Experience
Finalized	Report locked, audit trail visible, editor identity and timestamp shown
Network Error	Inform user the service could not be reached

11.5 Usability & Accessibility

- Clear, non-technical labels; main action always visible
- Draft text and finalized text are always visually distinguishable
- User-friendly, jargon-free error language
- Uploaded image and study context preserved through to the finalized report
- Sufficient contrast, keyboard-accessible controls, visible focus states, descriptive labels, alt text for meaningful images
- Responsive layout for desktop/laptop clinical workstations; no critical information conveyed by color alone

11.6 Responsible AI / Safety UX

Persistent notice: "AI-generated draft — not a final report until reviewed and finalized by a radiologist." The attention heatmap is labelled "Model Attention," never "Affected Area" or "Diagnosis Location." Design principles: Clarity · Traceability · Editability · Transparency · Human Review.

SECTION 12

Architecture & Design

12.1 High-Level Architecture

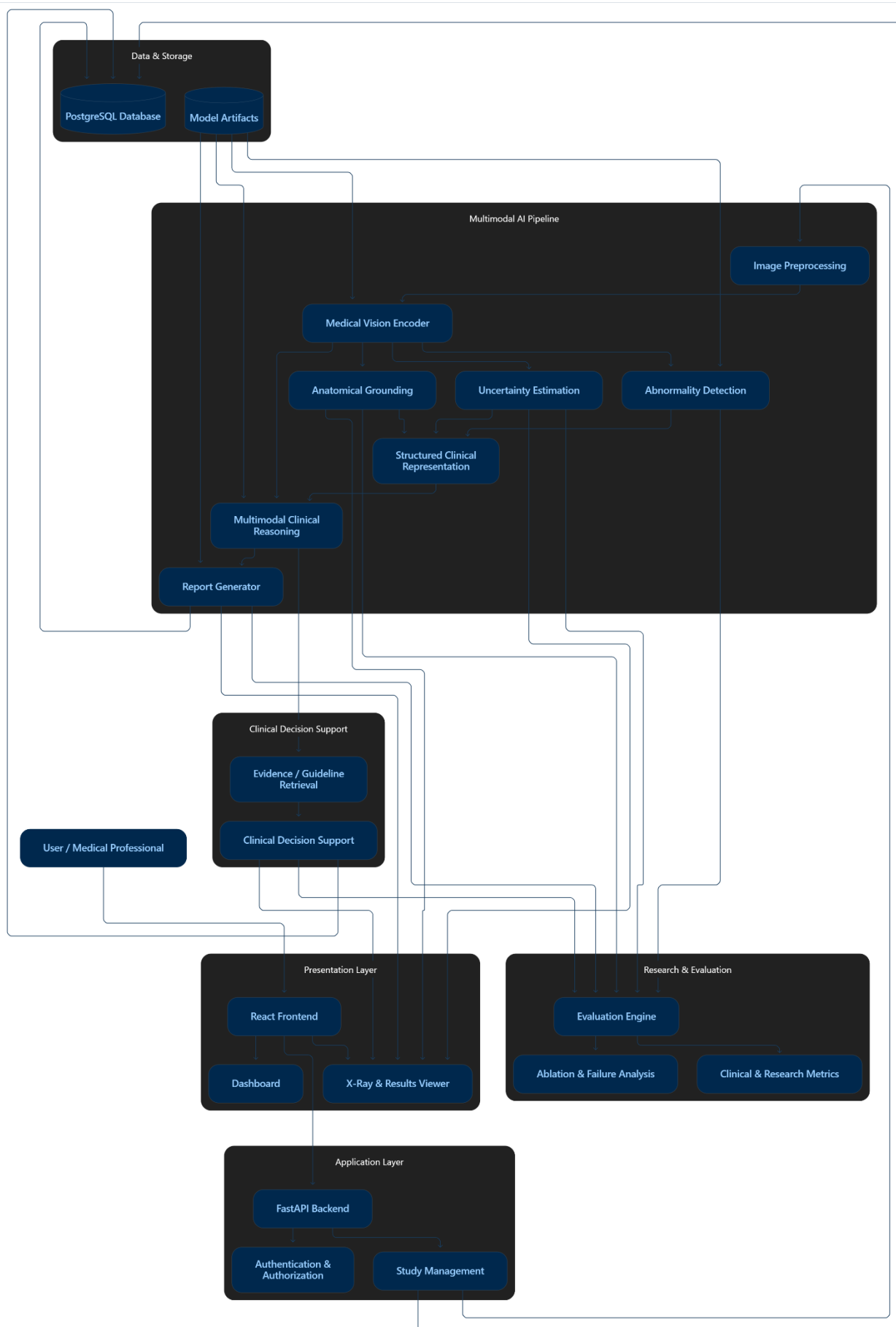
The proposed system follows a modular multimodal architecture designed to perform multiple stages of chest X-ray interpretation. The system consists of a presentation layer, application layer, multimodal AI pipeline, clinical decision-support layer, data storage, and research evaluation components.

The overall processing flow is:

Chest X-Ray → Image Preprocessing → Vision Encoder → Abnormality Detection → Anatomical Grounding → Structured Clinical Representation → Multimodal Clinical Reasoning → Report Generation → Clinical Decision Support

The architecture also incorporates uncertainty estimation and consistency analysis to identify unreliable or unsupported predictions.

12.1.1 High-Level Architecture Diagram



12.2 System Architecture Layers

The system is divided into the following layers:

- 1. Presentation Layer**
Provides the user interface for uploading X-rays, viewing AI analysis, inspecting anatomical grounding, reviewing generated reports, and viewing clinical decision-support outputs.
- 2. Application Layer**
Provides authentication, study management, API routing, inference orchestration, and communication between the frontend and AI services.
- 3. Multimodal AI Layer**
Performs image preprocessing, visual feature extraction, abnormality detection, anatomical grounding, uncertainty estimation, structured clinical representation, reasoning, and report generation.
- 4. Clinical Decision-Support Layer**
Uses structured findings and retrieved clinical evidence to generate diagnostic considerations, risk assessment, and management-oriented recommendations.
- 5. Data Layer**
Stores study information, generated results, model versions, evaluation results, and audit information.
- 6. Research & Evaluation Layer**
Evaluates diagnostic performance, report quality, grounding, uncertainty, hallucination, and model comparisons.

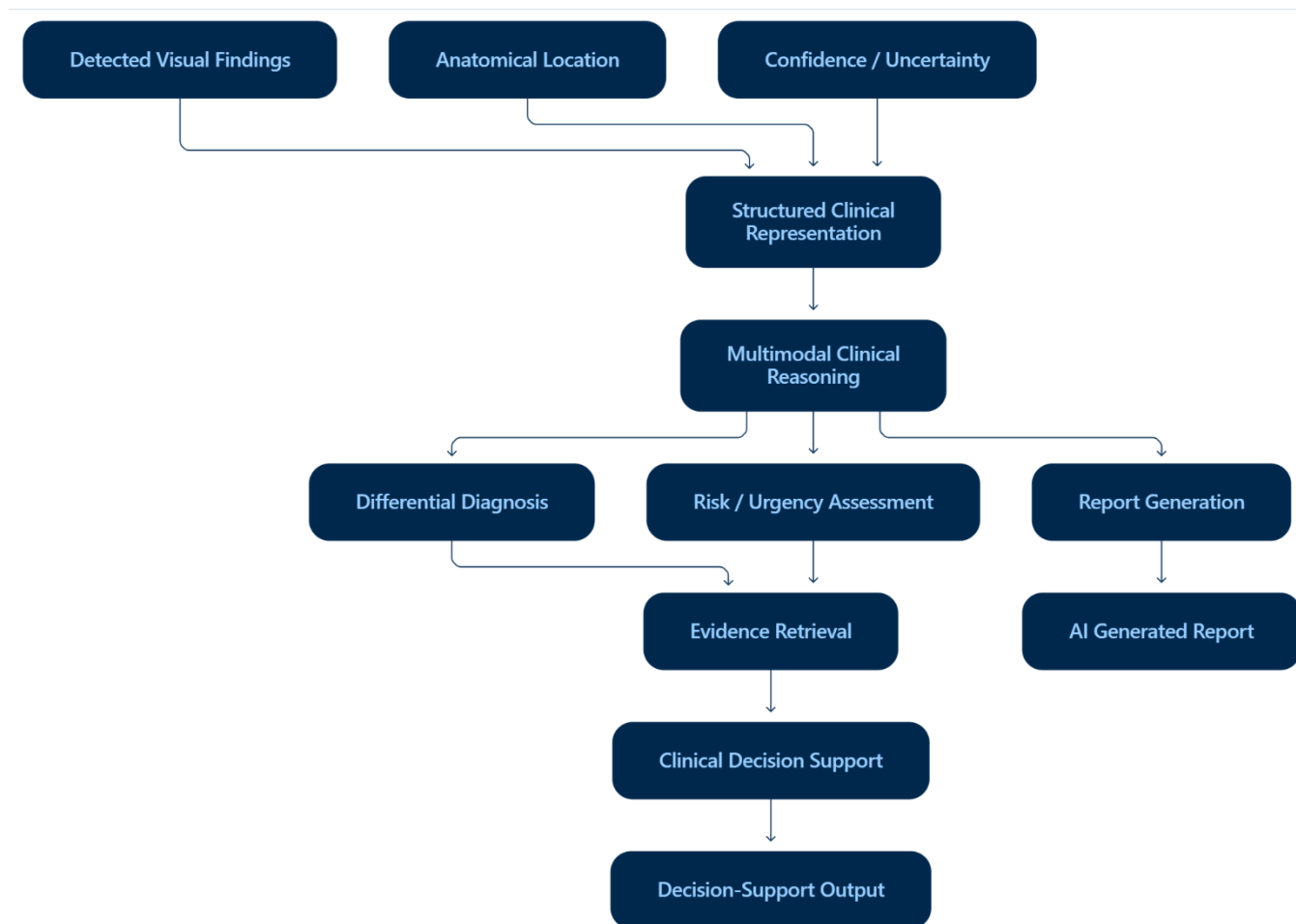
12.3 Chest X-Ray Analysis Flow

The detailed inference workflow is:



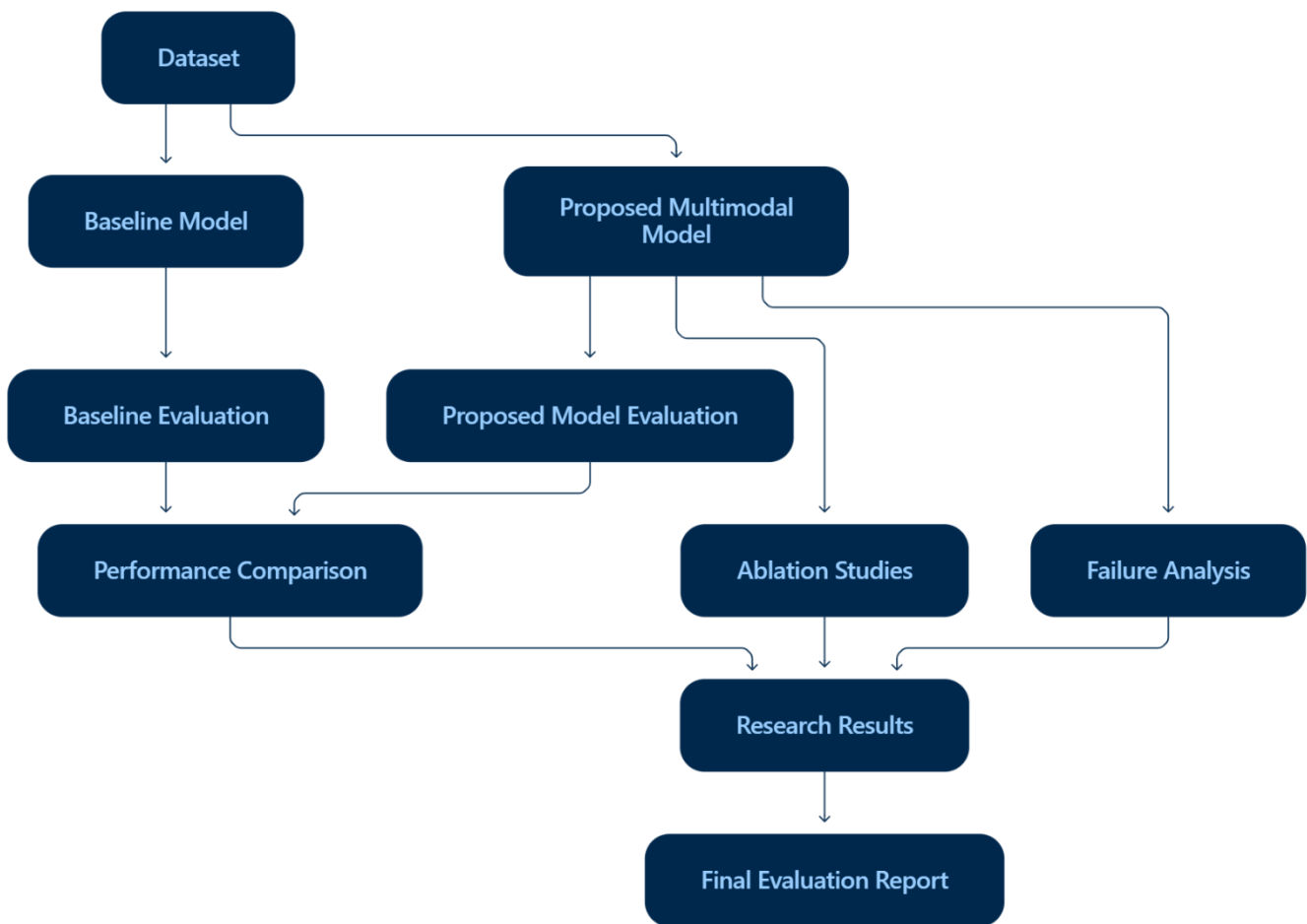
12.4 Clinical Reasoning Flow

The clinical reasoning component converts detected visual findings into structured clinical information before generating the report and decision-support output.



12.5 Research & Evaluation Architecture

The research module evaluates the proposed system against the CNN-LSTM baseline and performs component-level analysis.



12.6 Architecture Design Principles

The architecture follows these principles:

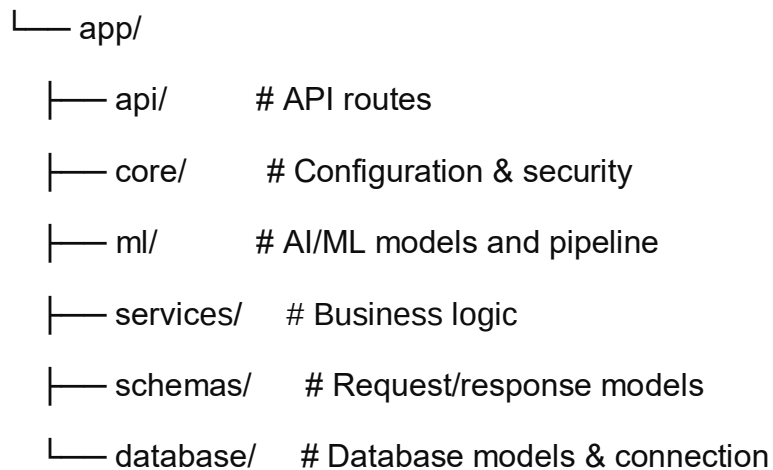
- **Modularity:** Each AI component performs a defined function and can be independently evaluated.
- **Grounding:** Generated findings should be linked to relevant visual evidence wherever possible.
- **Uncertainty Awareness:** The system should communicate uncertainty instead of forcing predictions.
- **Clinical Consistency:** Generated reports should remain consistent with detected findings.
- **Evidence-Based Reasoning:** Decision-support outputs should use relevant clinical evidence.
- **Research Reproducibility:** Model versions, experiments, metrics, and results should be tracked.

-
- **Extensibility:** The architecture should support future integration of additional patient information and clinical modalities.

SECTION 13 — LOW-LEVEL DESIGN (LLD)

13.1 Module Structure

backend/

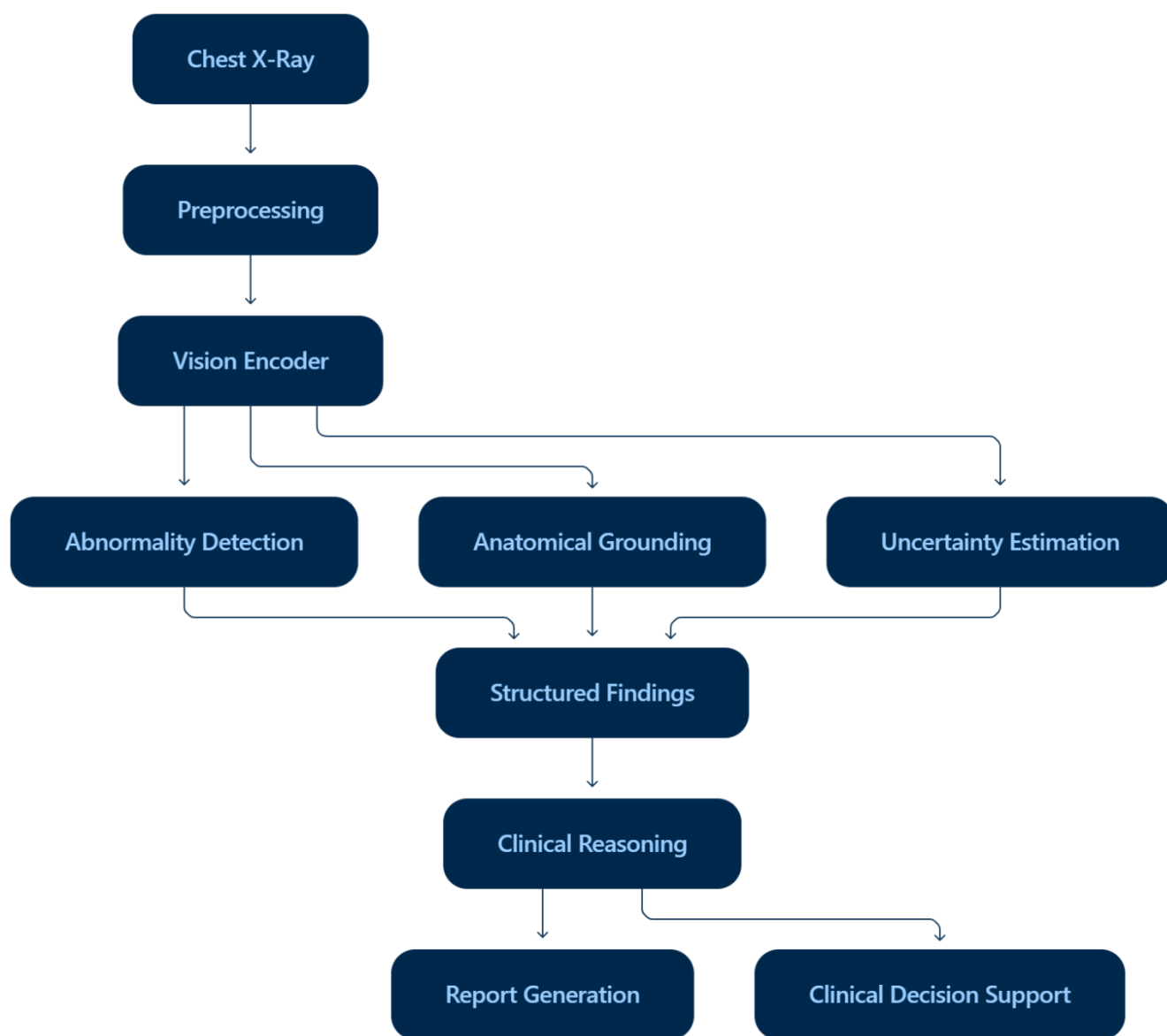


13.2 Core ML Modules

Module	Responsibility
Preprocessing	X-ray validation and preprocessing
Vision Encoder	Extract visual features
Abnormality Detector	Detect abnormalities
Grounding	Localize findings
Uncertainty	Estimate confidence

Module	Responsibility
Clinical Reasoning	Interpret structured findings
Report Generator	Generate Findings & Impression
CDS	Provide evidence-based decision support

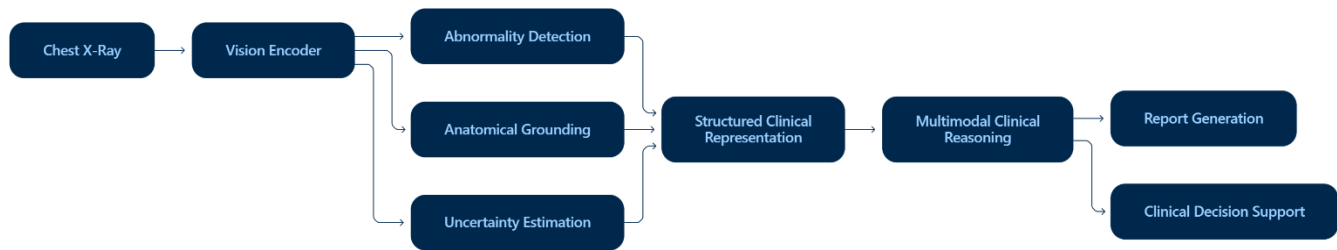
13.3 LLD Flow



SECTION 14 — ML / MULTIMODAL APPROACH

14.1 Proposed ML Architecture

The system uses a multimodal pipeline that converts chest X-ray images into structured clinical findings, radiology reports, and evidence-based clinical decision support.



14.2 Baseline vs Proposed Model

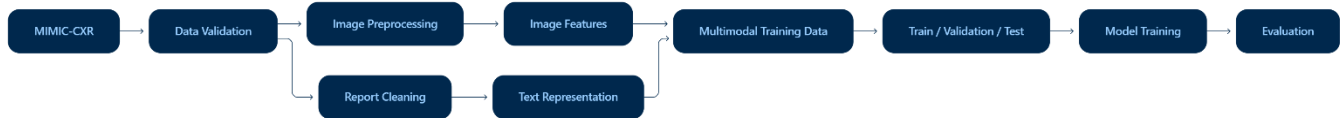
Component	Baseline	Proposed
Visual Encoder	CNN	Medical Vision Encoder
Decoder	LSTM	Multimodal Reasoning
Attention	Visual Attention Grounded Attention	
Abnormality Detection	Limited	Yes
Anatomical Grounding	No	Yes
Uncertainty	No	Yes
Clinical Reasoning	Limited	Yes
CDS	No	Yes

14.3 Research Approach

The proposed model will be compared with the CNN-LSTM + Visual Attention baseline using diagnostic, report-generation, grounding, factuality, and uncertainty metrics. Ablation studies will measure the contribution of individual components.

SECTION 15 — DATA PIPELINE

15.1 Data Processing Flow



15.2 Data Processing Steps

1. Load MIMIC-CXR using the official split.
2. Validate and preprocess chest X-ray images.
3. Clean and structure radiology reports.
4. Extract relevant clinical findings and labels.
5. Create image-text pairs.
6. Train and validate the proposed model.
7. Evaluate on the test set.

15.3 Data Split

Training: Model learning

Validation: Hyperparameter tuning and model selection

Testing: Final unbiased evaluation

SECTION 16 — CORE FEATURES

16.1 Core Features

Feature	Description
X-Ray Upload	Upload and validate chest X-ray images
Abnormality Detection	Identify clinically relevant abnormalities
Anatomical Grounding	Localize detected findings on the X-ray

Feature	Description
Uncertainty Estimation	Provide confidence for model predictions
Clinical Reasoning	Convert findings into structured interpretation
Report Generation	Generate Findings and Impression
Clinical Decision Support	Provide evidence-based diagnostic and management-oriented support
Explainability	Display findings, confidence and image grounding
Evaluation Dashboard	Compare model performance and research metrics

SECTION 17 — CLINICAL DECISION SUPPORT

17.1 Overview

The Clinical Decision Support (CDS) layer converts structured X-ray findings into evidence-based clinical support. It considers detected abnormalities, confidence, and retrieved medical evidence to generate diagnostic and management-oriented suggestions.



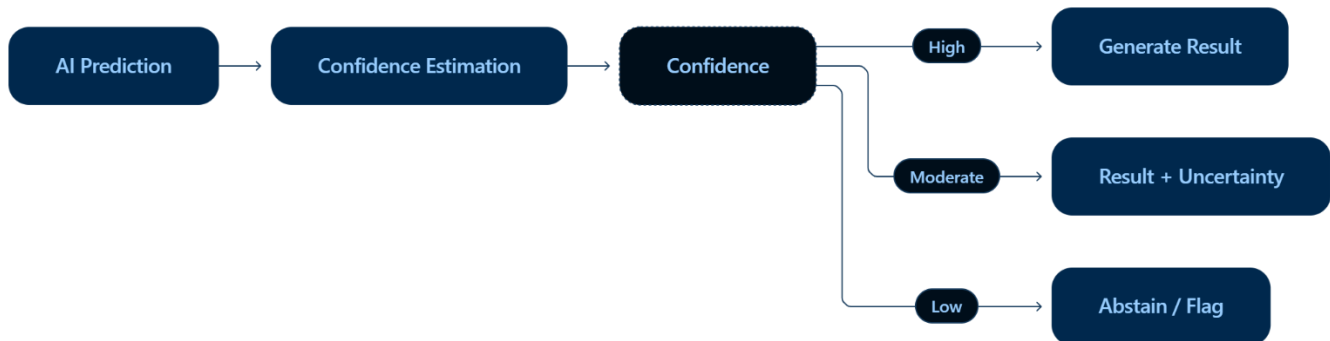
17.3 Key Requirements

- Evidence-grounded recommendations
- Clear confidence and uncertainty
- No unsupported clinical conclusions
- Traceability to supporting evidence
- Designed as clinical decision support, not autonomous prescribing

SECTION 18 — UNCERTAINTY, SAFETY & EXPLAINABILITY

18.1 Uncertainty & Safety

The system estimates confidence for detected abnormalities and generated outputs. Low-confidence or unsupported results are flagged to reduce overconfident conclusions.



18.2 Explainability

The system provides:

- Detected abnormalities
- Confidence/uncertainty
- Anatomical grounding
- Supporting evidence for CDS
- Clear indication of AI-generated content

18.3 Safety Requirements

- Detect unsupported findings and contradictions.
- Flag low-confidence outputs.
- Avoid presenting AI output as a definitive diagnosis.
- Keep human clinical review in the decision-support workflow.

SECTION 19 — NON-FUNCTIONAL REQUIREMENTS

Requirement	Description
Performance	Generate analysis within acceptable response time
Reliability	Provide consistent and reproducible outputs
Scalability	Support increasing studies and users
Security	Protect user data and system access
Explainability	Show confidence and grounding information
Maintainability	Use modular and well-documented components
Reproducibility	Track model versions, datasets and experiments
Usability	Provide a simple clinical-research interface

SECTION 20 — SECURITY & PRIVACY DESIGN

20.1 Security Requirements

- **JWT-based authentication and role-based access control.**
- **Secure API communication through HTTPS.**
- **Validate uploaded X-Ray, CT, MRI and Ultrasound files.**
- **Restrict access to studies, reports and clinical information.**
- **Secure storage of application data and AI outputs.**
- **Maintain audit logs for important system actions.**

20.2 Privacy Requirements

- Use de-identified datasets for research and development.
- Do not expose patient-identifiable information.
- Apply access controls to patient and study information.
- Maintain logs of access and system activity.
- Follow applicable dataset licenses and research requirements.

20.3 AI Safety

- Clearly label AI-generated outputs.
- Display confidence and uncertainty information.
- Do not allow autonomous diagnosis or prescribing.
- Require qualified human review before clinical use.

SECTION 21 — TESTING & EXPERIMENTAL DESIGN

21.1 Testing Strategy

The system will be evaluated at multiple levels:

- Input and image-quality validation
- Finding/disease detection
- Localization and segmentation
- Measurement accuracy
- Report factuality and clinical consistency

-
- Evidence grounding
 - Uncertainty calibration
 - RAG retrieval quality
 - Hallucination, omission and contradiction
 - Longitudinal comparison
 - End-to-end system performance

21.2 Experimental Comparison

Experiments will compare:

- CNN-LSTM baseline
- Proposed multimodal pipeline
- Individual component ablations
- Different model configurations where applicable

21.3 Failure Analysis

Failure cases will be categorized into:

- False positive
- False negative
- Incorrect localization
- Incorrect measurement
- Hallucinated statement
- Missing finding

- **Contradictory statement**
- **Unsupported recommendation**
- **Low-confidence prediction**

21.4 Human Review

Generated reports and clinical decision-support outputs will be reviewed by qualified reviewers where available. The system will remain a research prototype and human oversight will be required.

SECTION 22 — EVALUATION METRICS & SUCCESS CRITERIA

22.1 Evaluation Metrics

Category	Metrics
--- ---	
Finding Detection	AUROC, AUPRC, Precision, Recall, F1
Localization	IoU, Dice Score, Localization Accuracy
Segmentation	Dice, IoU
Measurement	MAE, RMSE
Report Generation	BLEU, ROUGE, BERTScore
Clinical Factuality	CheXbert F1, RadGraph F1
Grounding	Grounding Accuracy, Evidence Alignment
Uncertainty	Calibration Error, Brier Score
RAG	Retrieval Precision, Recall, Relevance
Safety	Hallucination, Omission, Contradiction Rate

22.2 Success Criteria

The proposed system should demonstrate measurable improvement over the CNN-LSTM baseline while maintaining grounded, traceable and appropriately uncertain outputs.

SECTION 23 — ABLATION STUDY & FAILURE ANALYSIS

23.1 Ablation Study

Evaluate the contribution of major components by removing them individually:



23.2 Failure Analysis

Analyze cases involving:

- False positives
- False negatives
- Hallucinated findings
- Missing findings

-
- **Incorrect localization**
 - **Overconfident predictions**
 - **Contradictory reports**
-

SECTION 24 — CI/CD AND OBSERVABILITY

24.1 CI/CD

- **Automated code testing**
- **API and integration testing**
- **Frontend testing**
- **Model version tracking**
- **Reproducible training configurations**
- **Automated build and deployment checks**

24.2 Observability

Monitor:

- **API health**
 - **Inference latency**
 - **Model failures**
 - **Prediction failures**
 - **Model version**
 - **Inference status**
 - **Evaluation results**
 - **System errors**
 - **Audit events**
-

SECTION 25 — LIMITATIONS & ETHICAL CONSIDERATIONS

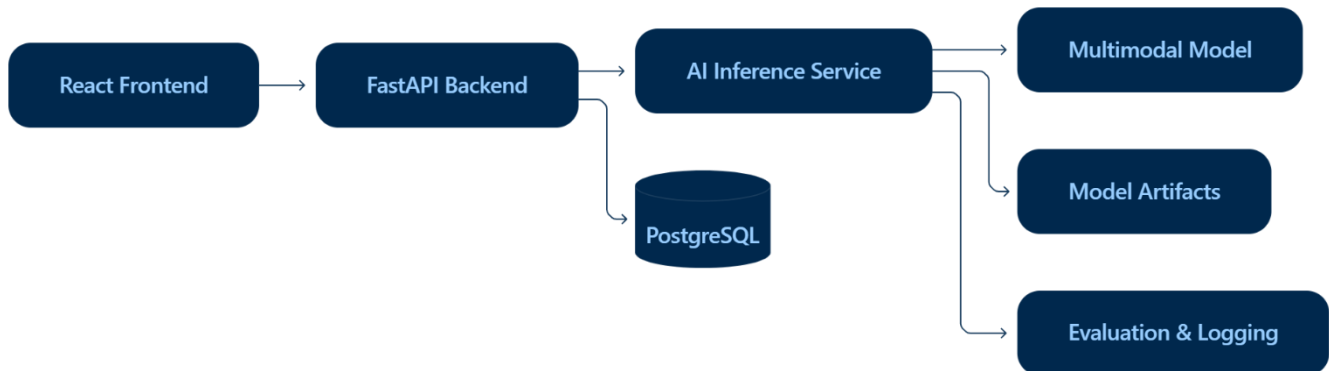
25.1 Limitations

- MIMIC-CXR does not represent all clinical populations.
- Modality-specific datasets may differ in size and annotation quality.
- Model performance may vary across institutions and imaging conditions.
- AI-generated reports may contain hallucinations or omissions.
- Grounding and confidence estimates may not always be clinically reliable.
- CT, MRI and Ultrasound may require additional specialized datasets and models.
- Clinical decision-support outputs require human validation.

25.2 Ethical Considerations

- Protect patient privacy and de-identified data.
- Clearly communicate AI uncertainty.
- Avoid unsupported clinical recommendations.
- Maintain human oversight.
- Evaluate model bias and performance across relevant populations.
- Do not use the system for autonomous diagnosis or prescribing.

SECTION 26 — DEPLOYMENT ARCHITECTURE



26.1 Deployment Components

- **Frontend:** React
- **Backend:** FastAPI
- **AI:** PyTorch
- **Database:** PostgreSQL
- **Model Storage:** Versioned model artifacts
- **Deployment:** Containerized research environment

SECTION 27 — REPOSITORY STRUCTURE & README

27.1 Repository Structure

project/

├─ backend/

| └─ app/

| | └─ api/

| | └─ core/

```
| | | — models/
| | | — services/
| | | — schemas/
| | — requirements.txt
| — Dockerfile
|
| — frontend/
| | — src/
| | — components/
| | — pages/
| | — package.json
|
| — models/
| | — xray/
| | — ct/
| | — mri/
| | — ultrasound/
|
| — data/
| — experiments/
| — tests/
| — docs/
| — README.md
```

27.2 README Requirements

The README shall include:

- Project overview
- Setup instructions
- Dataset information
- Model architecture
- API usage
- System architecture
- Evaluation instructions
- Research results
- Limitations
- Safety disclaimer

SECTION 28 — TEAM ROLES & COST ESTIMATION

28.1 Team Roles

Role Responsibility

--- ---

ML Developer Model development, training and evaluation

Backend Developer APIs, inference services and database

Frontend Developer Medical workstation UI and visualization

Research Lead Experiments, evaluation and documentation

28.2 Cost

The project primarily uses open-source software and research datasets.

Potential costs include:

- GPU / cloud computing
 - Model storage
 - Deployment infrastructure
 - Dataset-specific access requirements where applicable
-

SECTION 29 — TIMELINE

Phase	Work
--- ---	
Weeks 1–2	Dataset preparation and CNN-LSTM baseline
Weeks 3–4	Modality-specific vision models and finding detection
Weeks 5–6	Grounding, segmentation and uncertainty
Weeks 7–8	Structured clinical representation and report generation
Weeks 9–10	RAG, clinical reasoning and CDS
Week 11	Ablation and failure analysis
Week 12	Integration, documentation and final evaluation

SECTION 30 — ADRs & TRACEABILITY

30.1 Key Architecture Decisions

ADR	Decision	Reason
-----	----------	--------

|---|---|---|

| ADR-01 | CNN-LSTM baseline | Establish research benchmark |

| ADR-02 | Modality-specific models | Handle differences between imaging modalities |

| ADR-03 | Structured clinical representation | Connect image findings with reasoning and reports |

| ADR-04 | Evidence Chain | Improve traceability and grounding |

| ADR-05 | RAG-based evidence retrieval | Support evidence-grounded CDS |

| ADR-06 | Uncertainty estimation | Improve safety and confidence awareness |

| ADR-07 | Human review | Maintain clinical oversight |

30.2 Requirement Traceability

Requirements → Design → Implementation → Experiments → Evaluation

Each major research objective should be traceable to an implemented component and measurable evaluation result.

SECTION 31 — FINAL DELIVERABLES

31.1 Final Deliverables

1. MEDIVISTA AI multimodal research prototype
2. CNN-LSTM Chest X-Ray baseline
3. X-Ray, CT, MRI and Ultrasound analysis pipeline
4. Finding detection and anatomical grounding
5. Localization / segmentation where applicable

-
6. Structured clinical representation
 7. Grounded Findings and Impression report generation
 8. Evidence Chain
 9. Uncertainty and safety mechanisms
 10. RAG-based Evidence Library
 11. Clinical Decision Support module
 12. Longitudinal comparison module
 13. React + FastAPI application
 14. Model Evaluation dashboard
 15. Failure Analysis module
 16. Model Registry
 17. Human Review & Approval workflow
 18. Evaluation and ablation results
 19. Technical documentation
 20. Final research report / paper

31.2 Safety Boundary

The final system will remain an academic research prototype. It will provide AI-assisted analysis and decision-support information but will not autonomously diagnose, prescribe medication, make final treatment decisions or replace qualified healthcare professionals.